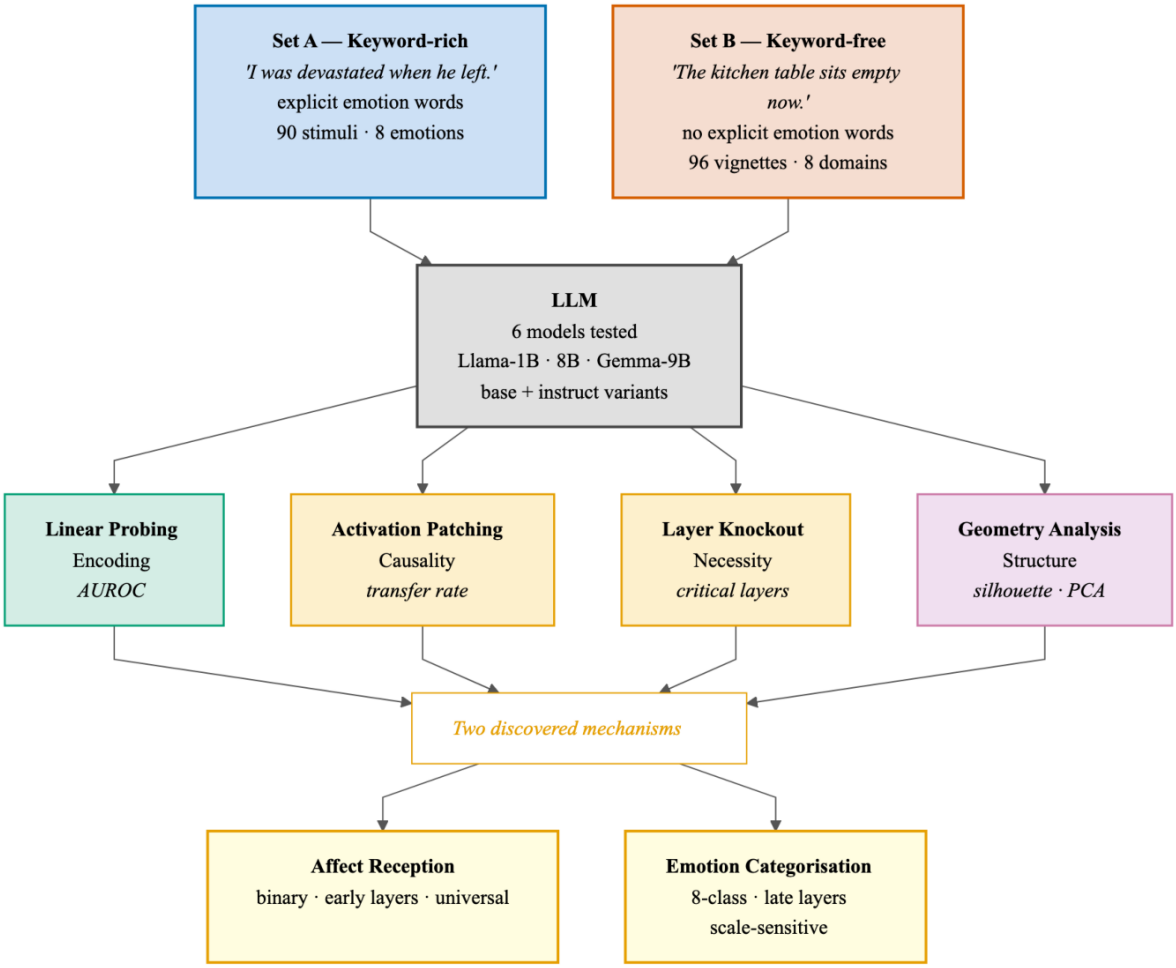

Figure 1: Experimental design overview. Both keyword-rich stimuli (Set A; explicit emotion vocabulary) and clinical vignettes (Set B; situational cues, no emotion words) are processed by the same six LLMs. Four convergent analysis methods—linear probing, activation patching, knockout experiments, and representational geometry—probe internal activations at every layer, yielding evidence for two dissociable mechanisms: *affect reception* (binary, early-layer, universal) and *emotion categorisation* (8-class, mid-to-late layer, scale-sensitive).

2 Related Work

2.1 Emotion Circuits in LLMs

The mechanistic study of emotion processing in LLMs is recent but growing rapidly. Tak et al. [1] provide the most comprehensive treatment, demonstrating that emotion is linearly decodable from mid-layer activations, causally transferable via activation patching, and steerable using emotion direction vectors. Their study spans multiple model families and sizes, establishing mid-layer consolidation as a robust phenomenon. We treat their work as our primary replication target: we first reproduce their methodology with their stimulus type, then extend it with clinical stimuli that test whether their findings survive keyword removal.

Lee et al. [2] ask "Do LLMs Have Emotion Neurons?" and identify neurons that fire selectively on emotional text. Their discussion section is, in effect, our research question—they note that identified neurons may respond to cohyponyms or syntactic frames rather than emotional

content, but do not test this possibility. We operationalize their concern as an experimental contrast.

Reichman et al. [3] map an "emotional manifold"—a low-dimensional subspace of model activations organized by emotion—and demonstrate that learned intervention modules can steer emotion perception along these directions. Their finding that emotion directions generalize across datasets and languages is suggestive of genuine emotional representation, but their stimuli share the same keyword confound.

Zhang and Zhong [8] examine emotion encoding depth in a systematic layer-wise probing study, showing that emotional signals form clusters in representation space that are sharper internally than what zero-shot output classification reveals. Their finding that emotional information persists over hundreds of subsequent tokens is consistent with the affect reception mechanism we report here.

Additional evidence for structured internal emotion geometry comes from work on neural emotion decoding [9], circuit-level emotion manipulation [4], and emotion-sensitive neurons in audio-language models [10], extending the picture across modalities and methodologies.

Beyond emotion specifically, Chen et al. [11] identify linear directions in LLM activation space underlying character traits including sycophancy, hallucination propensity, and ethical alignment—*persona vectors*. Their finding that these directions predict and track personality shifts after finetuning, and can be used to monitor trait fluctuations at deployment time, establishes a broader paradigm: psychological properties of language models are linearly represented in activation space and causally relevant to behavior. Our work on emotion processing operates within this paradigm, extending it with a clinical validity criterion that prior work—on both emotion and character—has not applied.

At the theoretical level, the Persona Selection Model [12] proposes that LLMs learn to simulate diverse personas during pre-training, with emotional representations among the character-consistent features installed across the training distribution. This framework generates two testable predictions relevant to our study: that emotional encoding should be a pre-training phenomenon—observable in base models before any alignment training—and that psychological methods designed for assessing human emotional processing should transfer productively to AI internal representations. We test both predictions directly (sections 8.3 and 9.6).

2.2 The Keyword Confound as a Clinical Validity Problem

All existing studies use stimuli drawn from crowd-enVENT [5], GoEmotions [6], DailyDialog [7], or synthetic prompts containing explicit emotion vocabulary. This is not a minor methodological quibble—it is a clinical validity problem. In psychology, a measure of emotional processing must demonstrate that it responds to emotional *meaning*, not to the surface markers that correlate with emotional meaning. A depression screener that only detects patients who say "I am depressed" has no clinical validity. The same standard applies to internal-representation claims about LLMs: can a mechanistic finding survive removal of lexical markers while preserving ecological emotional meaning? If not, the claim describes keyword detection, not emotion processing.

The confound is acknowledged but untested. What is needed is straightforward: stimuli that carry emotional meaning without carrying emotion vocabulary. If probes still decode emotion from these, the keyword-spotting explanation fails. If they do not, the existing claims need revision. Separately, Schaeffer et al. [13] demonstrate that metric choice alone can create illusory discontinuities in the emergent abilities literature—a cautionary example of how measurement artifacts masquerade as genuine capabilities. We follow their recommendation to use continuous metrics (AUROC rather than accuracy) throughout.

2.3 Clinical Stimulus Methodology

Clinical psychology has spent decades developing rigorous methodologies for creating text-based stimuli that evoke specific emotional responses at controlled intensities. Validated verbal emotion vignettes [14] demonstrate that carefully constructed situational narratives reliably elicit targeted emotions through context rather than explicit labeling, with hit rates above 80% and intensity ratings in the 65–90% range across anger, grief, and fear conditions. Crisis vignettes from the LEAP tradition [15] show that factorial vignette designs—varying one situational factor at a time—isolate emotional content from confounding contextual variables while maintaining ecological validity. PTSD script-driven imagery protocols [16, 17] establish that first-person situational narratives with no explicit emotional labeling reliably produce differential physiological and psychological responses across emotion categories. Grief narrative design has its own methodological literature, with established guidelines for constructing bereavement stimuli that elicit genuine emotional salience through situational description rather than internal state disclosure [18]. Vignette factorial methodology more broadly has been validated as a tool for isolating psychological constructs in controlled experimental designs [19].

To our knowledge, clinical stimulus methodology has not been applied to mechanistic analysis of internal model states. The tools exist in clinical psychology. The gap is in mechanistic interpretability, where nobody has used them.

2.4 Affect Reception in Affective Neuroscience

The term "affect reception" draws on established neuroscience of receptive emotional processing. The human brain processes emotionally charged input through rapid, parallel circuits: the "low road" subcortical pathway (thalamus $\rightarrow$ amygdala, $\sim$12 ms) provides crude valence and arousal tagging before the "high road" cortical pathway ($\sim$40 ms) enables categorical emotion identification [20, 21]. This produces a temporal dissociation: organisms detect "something emotionally significant is happening" before they identify what specific emotion is present. Affective neuroscience further distinguishes *receptive* emotional processing—the automatic detection and encoding of emotional significance in input—from *expressive* processing—the generation of emotional behavior and response—as separable systems with distinct neural substrates, timing, and modulating conditions [22, 23]. Early ERP components (P1/N170, $\sim$100–200 ms) index receptive salience detection independent of any motor or expressive response; categorical identification requires later cortical processing ($\sim$250–400 ms) [24].

We are not claiming that LLMs have amygdalas or that transformer layers map onto subcortical pathways. We use this framework as a *structural analogy* for naming the dissociation we